Class 2

- ❑ Ohm's Law and its Applications
- ❑ Energy and Power
- ❑ Resistor and Resistor Color Code
- ❑ Metric Prefixes

Content Courtesy:

Electronics Technology Fundamentals, 3rd ed. Paynter and Boydell

Electronics Fundamentals 9th edition Floyd/Buchla/Snyder

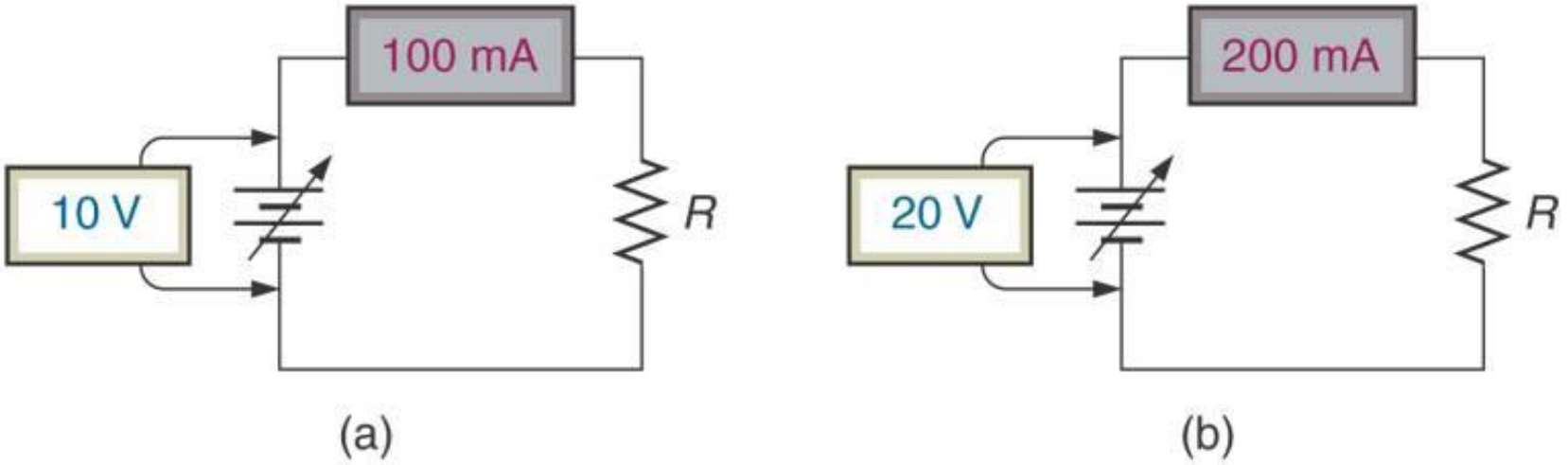
Ohm's Law

The most important fundamental law in electronics is **Ohm's law**, which relates voltage, current, and resistance.

- *German Physicist **George Simon Ohm** found that:*
  - ***Current** is **directly proportional** to **voltage***
  - ***Current** is **inversely proportional** to **resistance***

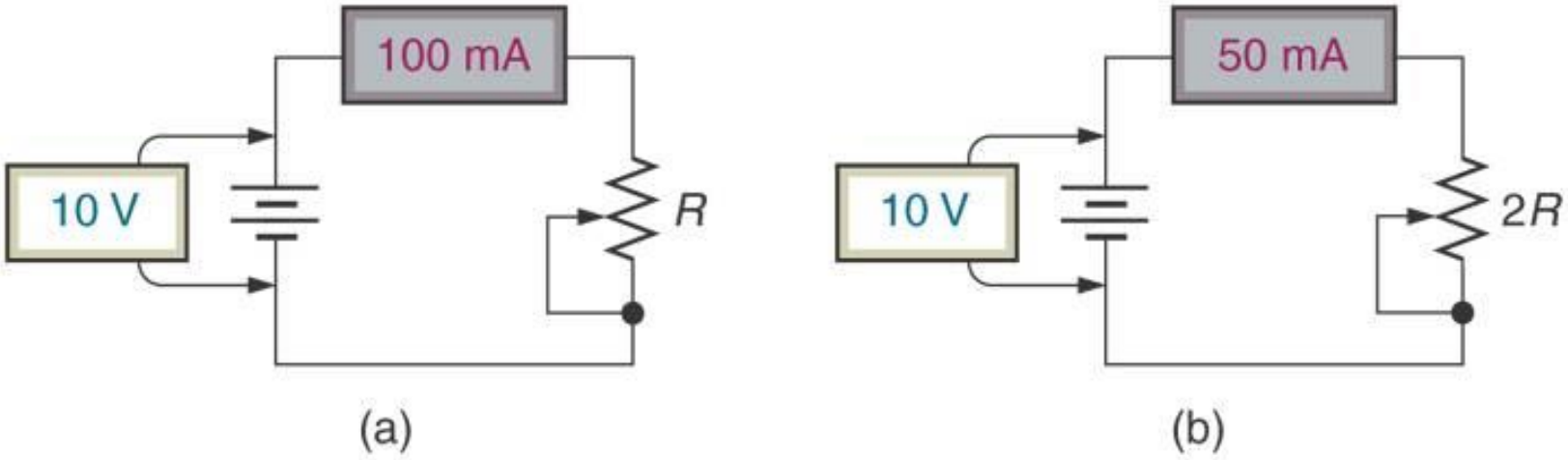
Ohm's Law

- *The **Relationship** Between **Current** and **Voltage***
  - *As voltage **increases**, current also **increases***



Ohm's Law

- *The Relationship Between* ***Current and Resistance***
  - *As* ***resistance*** *increases*, ***current*** *decreases*



Ohm's Law and Circuit Calculations

- Using Ohm's Law to Calculate Voltage

$$V = IR$$

I = the circuit current
V = the applied voltage
R = the circuit resistance

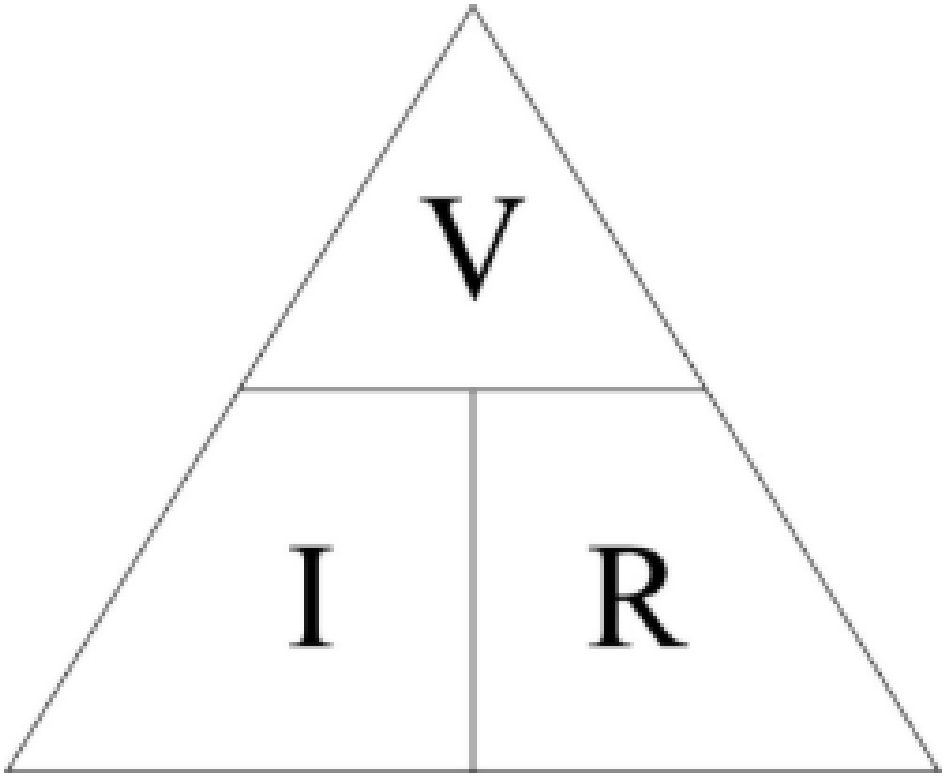
- Using Ohm's Law to Calculate Resistance

$$R = \frac{V}{I}$$

- Using Ohm's Law to Calculate Current

$$I = \frac{V}{R}$$

Ohm's Law Triangle



Ohms Law - Problems

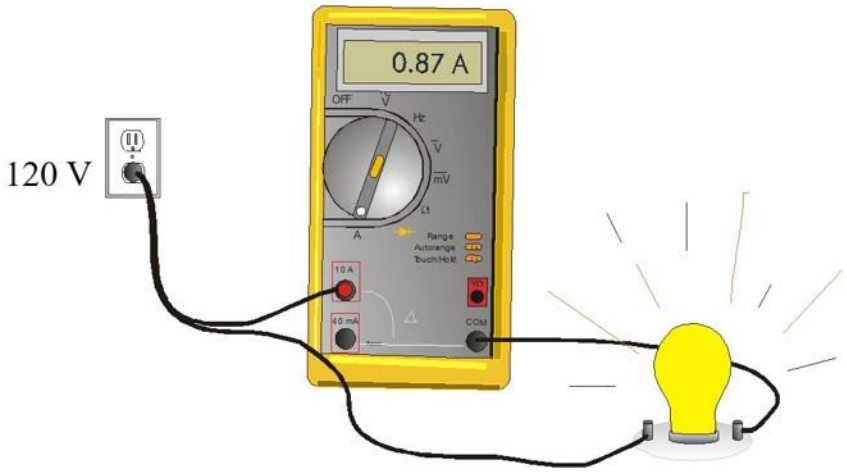
Q1. What is the voltage across a 680 Ω resistor if the current is 26.5 mA?

18.0 V

Q2. What is the voltage across a 1.5 MΩ resistor if the current is 4.0 μA?

6.0 V

Q. What is the (hot) resistance of the bulb? 138 Ω



Energy and Power

- **Energy** is closely related to work. Energy is the ability to do work. As such, it is measured in the same units as work, namely the newton-meter (N-m) or joule (J).

- **Power** – Power is the rate of doing work. The amount of energy used per unit time.
- Unit of Power: 1 watt (W) = 1 joule/second

$$P = \frac{W}{t}$$

P = the power used, in watts (W)
W = Energy in Jules (J)
t = Time in seconds (s)

One **watt** is the amount of power when one Joule of energy is used in one second

Energy and Power

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- **Power** – the amount of energy used per unit time
- Unit of Power: 1 **watt** (W) = 1 joule/second
- Calculating Power

$$P = VI$$

where

P = the power used, in watts (W)

V = the applied voltage, in volts (V)

I = the generated current, in amperes (A)

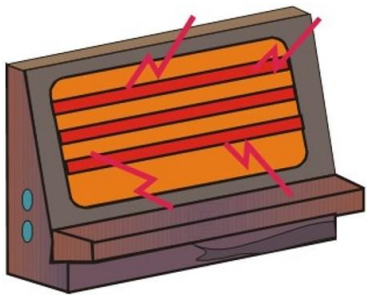
Power in Electric Circuit

In electrical work, the rate energy is dissipated can be determined from any of three forms of the power formula.

$$P = I^2R \qquad P = VI \qquad P = \frac{V^2}{R}$$

Together, the three forms are called **Watt's law.**

Q. What power is dissipated by a heater that draws 12 A of current from a 120 V supply?



$$P = IV$$
$$= (12\ \text{A})(120\ \text{V})$$
$$= 1440\ \text{W}$$

Problem:

What power is dissipated in a $27\,\Omega$ resistor is the current is 135 mA?

Solution:

Given that you know the resistance and current, substitute the values into $P = I^2R$.

$$P = I^2R$$

$$= (0.135\text{ A})^2 (27\,\Omega)$$

$$= 0.492\text{ W}$$

Kilowatt-Hour (kWh) Unit of Energy

- The kilowatt-hour (kWh) is a much **larger unit of energy** than the joule. There are 3.6×10^6 J in a kWh.
- The kWh is convenient for electrical appliances.
- **Kilowatt-hour (kWh)** is a unit of measurement for the amount of electricity a device uses over a period of time.
- It's defined as one kilowatt (1,000 watts) of power used for one hour.

$W = Pt = (1000W) * (1h) = 1kWh$

Q. What is the energy used in operating a 1200 W heater for 20 minutes?

$1200\ W = 1.2\ kW$

$20\ min = 1/3\ h$

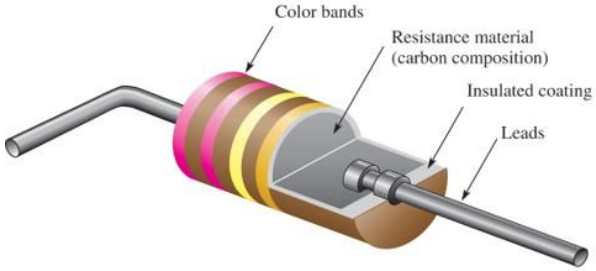
$1.2\ kW \times 1/3\ h = 0.4\ kWh$

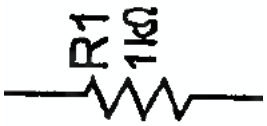
- Understand the usage of electricity and its cost: OhioEdison

Resistors

Components designed to have a specific amount of resistance are called *resistors*.

- Resistors
  - Limit current
  - Unit – Ohm (Ω)

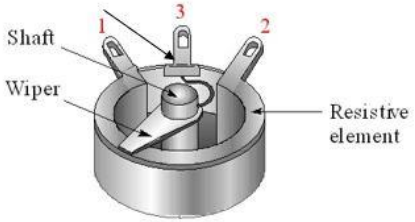


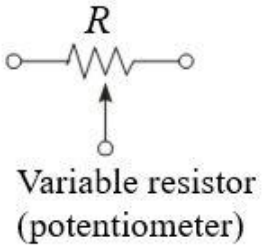


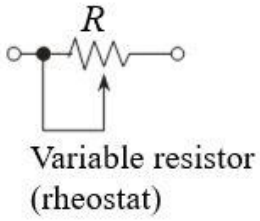
Fixed Value and Variable Value

- Variable resistors include the potentiometer and rheostat. The center terminal of a variable resistor is connected to the wiper.









Resistor Color Codes

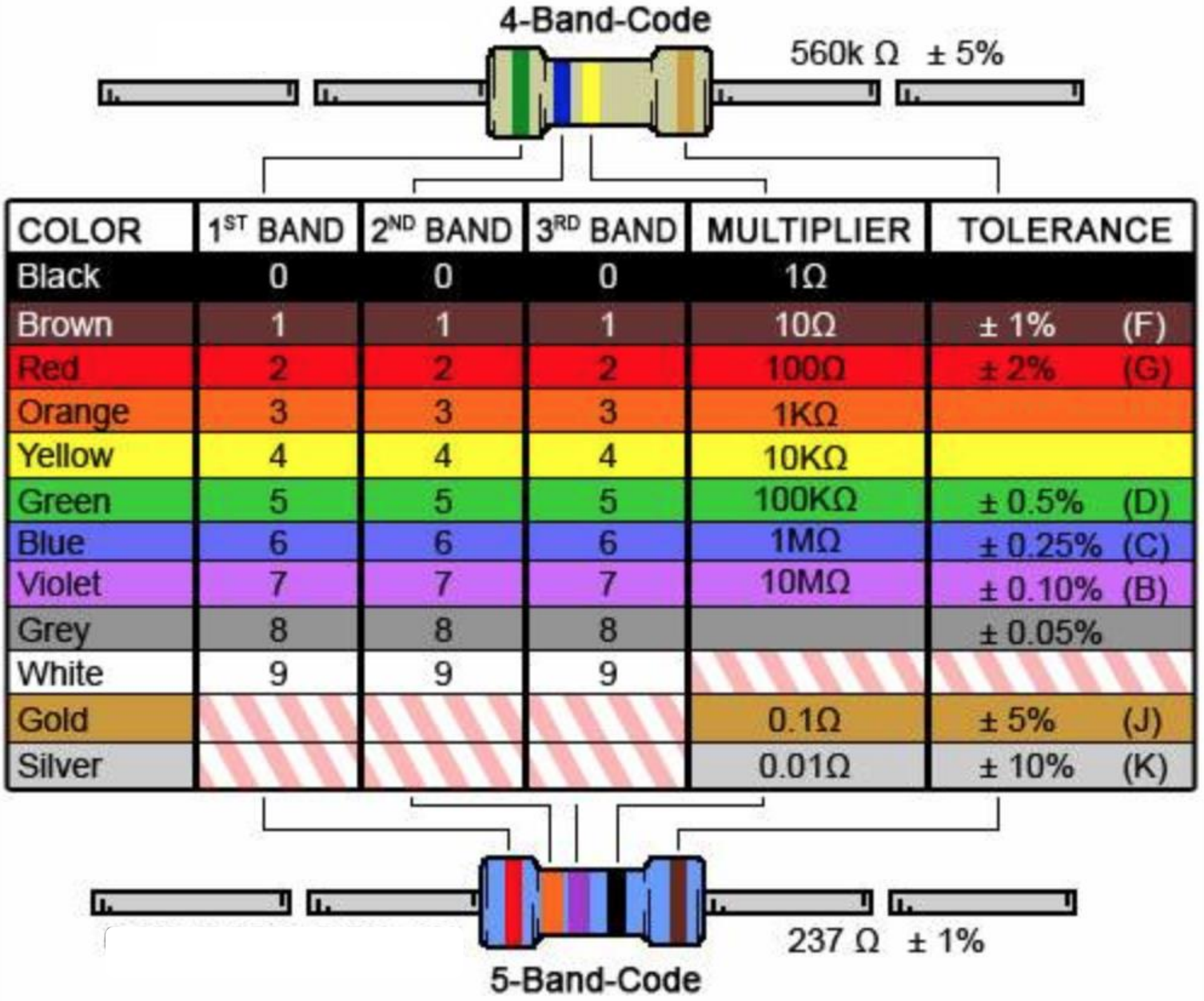
- The Color Code for Standard (Non precision) Resistors – Color bands indicate the resistance value
  - Usually 4 or 5 bands



Resistor Color Codes

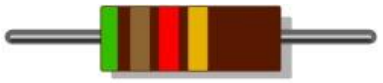
	Color		Digit	Multiplier	Tolerance
Resistance value, first three bands: First band – 1st digit Second band – 2nd digit *Third band – Multiplier (number of zeros following second digit)		Black	0	10^0	
		Brown	1	10^1	1% (five band)
		Red	2	10^2	2% (five band)
		Orange	3	10^3	
		Yellow	4	10^4	
		Green	5	10^5	
		Blue	6	10^6	
		Violet	7	10^7	
		Gray	8	10^8	
		White	9	10^9	
Fourth band – tolerance		Gold	±5%	10^{-1}	5% (four band)
		Silver	±10%	10^{-2}	10% (four band)
		No band	±20%		

* For resistance values less than 10 W, the third band is either gold or silver. Gold is for a multiplier of 0.1 and silver is for a multiplier of 0.01.



COLOR	1ST BAND	2ND BAND	3RD BAND	MULTIPLIER	TOLERANCE
Black	0	0	0	1Ω	
Brown	1	1	1	10Ω	± 1% (F)
Red	2	2	2	100Ω	± 2% (G)
Orange	3	3	3	1KΩ	
Yellow	4	4	4	10KΩ	
Green	5	5	5	100KΩ	± 0.5% (D)
Blue	6	6	6	1MΩ	± 0.25% (C)
Violet	7	7	7	10MΩ	± 0.10% (B)
Grey	8	8	8		± 0.05%
White	9	9	9		
Gold				0.1Ω	± 5% (J)
Silver				0.01Ω	± 10% (K)

Courtesy: Digikey.com



5.1kΩ ± 5%



820 kΩ ± 10%



47 Ω ± 10%



1.0 Ω ± 5%

Wire Resistance

Sometimes, the resistance of wires must be accounted for.

The equation for wire resistance is: $R = \frac{\rho l}{A}$

$$\text{where } \rho = \text{resistivity in CM} - \Omega/\text{ft}$$
$$l = \text{length in feet}$$
$$A = \text{cross sectional area in circular mils (CM)}$$

Example

What is the resistance of 400 feet of 22 gage copper wire? The area is 642 CM and the resistivity of copper is 10.37 CM-Ω/ft.

✓By the equation, $R = \frac{\rho l}{A} = \frac{(10.37\,\Omega\text{-CM/ft})(400\text{ ft})}{642\text{ CM}} = 6.46\ \Omega$

Engineering Metric Prefixes

Large

Can you name the prefixes and their meaning?

P	**peta**	10^{15}
T	**tera**	10^{12}
G	**giga**	10^{9}
M	**mega**	10^{6}
k	**kilo**	10^{3}

Engineering Metric Prefixes

Small

Can you name the prefixes and their meaning?

m	**milli**	10^{-3}
μ	**micro**	10^{-6}
n	**nano**	10^{-9}
p	**pico**	10^{-12}
f	**femto**	10^{-15}

Metric Systems & Conversions

Engineering Notation

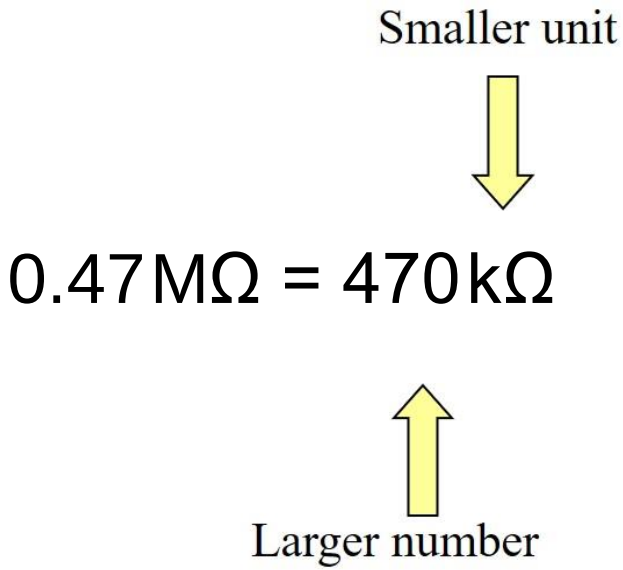
- Scientific notation that uses standard prefixes

Range	*Values*	*Prefix*	*Symbol*
(1 to 999) x 10^{12}	(1 to 999) trillion	tera-	T
(1 to 999) x 10^{9}	(1 to 999) billion	giga-	G
(1 to 999) x 10^{6}	(1 to 999) million	mega-	M
(1 to 999) x 10^{3}	(1 to 999) thousand	kilo-	k
(1 to 999) x 10^{0}	1 to 999	(none)	(none)
(1 to 999) x 10^{-3}	(1 to 999)	milli-	m
(1 to 999) x 10^{-6}	thousandths	micro-	μ
(1 to 999) x 10^{-9}	(1 to 999) millionths	nano-	n
(1 to 999) x 10^{-12}	(1 to 999) billionths (1 to 999) trillionths	pico-	p

Metric Conversions

When converting from a larger unit to a smaller unit, move the decimal point to the right. Remember, a smaller unit means the number must be larger.

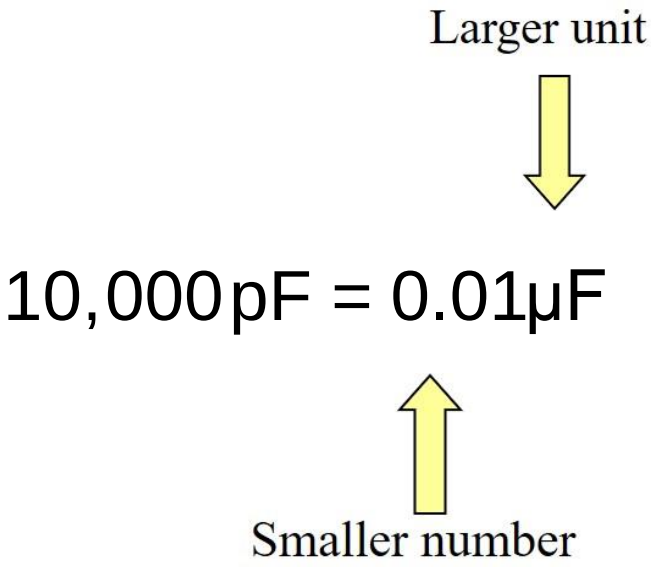
Example 1



Metric Conversions

When converting from a smaller unit to a larger unit, move the decimal point to the left. Remember, a larger unit means the number must be smaller.

Example 2



Metric Arithmetic

When adding or subtracting numbers with a metric prefix, convert them to the same prefix first.

Example 1 $10{,}000\ \Omega + 22\ \text{k}\Omega\ =$

$$10{,}000\ \Omega + 22{,}000\ \Omega = 32{,}000\ \Omega$$

Alternatively,

$$10\ \text{k}\Omega + 22\ \text{k}\Omega\ = 32\ \text{k}\Omega$$

Metric Arithmetic

When adding or subtracting numbers with a metric prefix, convert them to the same prefix first.

Example 2 $200\ \mu A + 1.0\ mA =$

$$200\ \mu A + 1{,}000\ \mu A = 1{,}200\ \mu A$$

Alternatively,

$$0.200\ mA + 1.0\ mA = 1.2\ mA$$